

Gerrit Geeraerts

Bridging the gap between today's programming and tomorrow's AI, while deeply rooted in the open-source community's collaborative spirit.

 LinkedIn |  ggeeraer@gmail.com |  GitHub |  Stack Overflow |  Ask Ubuntu |  Driving License

About Me

As a passionate tech enthusiast, I am driven by the belief that technology is a powerful tool for enhancing efficiency and productivity.

As a person, I am a collaborative team player with a passion for continuous learning and knowledge sharing. Motivated, a creative problem-solver, leveraging critical thinking and analytical skills to find robust solutions. I am also a generalist with a broad interest range. My goal is to find a team with like-minded people and continuously expand our technical expertise by learning and integrating the latest technologies to build robust solutions that enhance our productivity and efficiency.

Experience

2024

AI Engineer Intern, Cronos (SUMSUM)

As part of my becode training I completed my 3-month internship at Cronos (SUMSUM) with a focus on Large Language Models

AI-Powered Sales Enablement Chatbot: Developed a Retrieval-Augmented Generation (RAG) chatbot designed to let businesses experience the capabilities of AI by interacting with their own business data, websites, and uploaded documents. The chatbot enabled SUMSUM's technical team to set up demo's in minutes and the sales team to deliver rapid demos, creating a 'wow effect' with the client. Cost optimizations were implemented to run multiple demos parallel, Link: AI-Powered Sales Enablement Chatbot

Project Manager Assistant Chatbot: Designed and developed an AI-powered Project Manager Assistant chatbot leveraging a collaborative multi-agent system to enhance client communication. The solution enables project managers to ask complex, context-specific queries, with the agents working together to retrieve relevant data, synthesize insights, and dynamically generate reports enriched with visual elements such as charts and tables. This reduces client inquiry response times by automating manual research and delivering real-time, data-driven reports., Link: Project Manager Assistant Chatbot

Skills: **Python, Git, Docker, Linux, Bash**, AWS EC2, AWS S3, AWS Opensearch, Streamlit, LLM, LLM Agents, Langchain, Langgraph, Langsmith, RAG, Ragas, GPT, Claude, Bash, LLM

2024

Data Engineer Trainee, Becode

A full-time 6-month AI training with a focus on data engineering, we worked mostly on projects

Credit Card Churning: Designed and deployed an Azure ML infrastructure using Terraform to predict customer credit card churn. Automated provisioning of cloud resources and integrated with ML Flow for experiment tracking and model management. Leveraged CatBoost for churn prediction, implemented MLOps best practices for seamless workflows., Link: Credit Card Churning

Immo Price Prediction App: Developed and deployed a housing price prediction app utilizing a machine learning model. Hosted the prediction model on a FastAPI-based backend. Designed a user-friendly front-end with Streamlit, enabling interaction with the model through an intuitive interface., Link: Immo Price Prediction App

Housing Price Prediction with ML: Machine learning project focused on predicting housing prices by building a robust data pipeline and thoroughly exploring and optimizing various models. Feature engineering, Hyperparameter tuning, exploring different models (including Linear Regression, Random Forest, and CatBoost), Link: Housing Price Prediction with ML

Wine Market Analysis: Collaboratively built an OLAP database from an OLTP source to power a Streamlit dashboard, enabling wine trend analysis, top wine selection., Link: Wine Market Analysis

ArcelorMittal Job Fair Chatbot: Developed a chatbot leveraging a Scrapy scraper to extract data from ArcelorMittal's website and job database. The bot starts out by encouraging the user to ask questions about the company, and then nudges the user into sharing interest and skills to match with job openings. In the background the bot uses vector similarity to match the user's skills with the job openings., Link: ArcelorMittal Job Fair Chatbot

Skills: **Python, API, ML Ops, Docker, Kubernetes, Kanban, Scrum, Linux, FastAPI, SQL**, ML Flow, Azure ML, Scikit Learn

2024

System Engineer, Viso Cor Mariae

Volunteered to analyze IT infrastructure, create a roadmap, and start on implementing

Skills: Migration to Azure Cloud, **Powershell Programming**

2019-2023 **System Engineer / Developer**, Kasterlinden

Worked on digitalizing education and managing IT infrastructure.

Mobile Device Management: Introduction of Ipads in the organisation, research possible mdm solutions and implementation. Maintenance friendly solution with automatic synchronizing accounts and groups with existing systems. Barcodes system for fast inventory management and reallocating iPads. Extending Wi-Fi infrastructure.

Cloud Migration: Creating a more maintenance friendly environment by merging and migrating multiple domains, subnets and servers to the (Azure) cloud, with a minimal impact on the end user. Moving DNS, DHCP to Firewall, moving local ad, file and print server to the cloud.

Application Self-Service: Created a tool to streamline the creation of Intune packages, enabling users to install software through a self-service portal. Together with Delivery Optimization to optimize Network Load.

Modern Inventory Management: Developed an automated inventory system that hooked into Lansweeper for improving asset management and oversight.

Give Trainings: Motivate and train teachers to work with modern education tools like iPads.

Network Management: Collaborated closely with our parent organization to manage our Wi-Fi, LAN, WAN, firewalls.

Skills: **Django, Python, Git, Docker, SQL, Cloud Migration, Azure Cloud, Selenium, Networking: (DHCP, DNS, Exchange, Firewall), Powershell Programming, HTML, CSS, Azure Cloud**

2019-Present **Developer**, Home Projects

Various personal projects involving software development and automation.

LLM Search Assistant: A multi-agent system that emulates human browsing on a website to find an answer to a user question. Integrated in OpenWebUI with pipelines and FastAPI.

Crypto Bots: Created a backtesting framework for quick testing of algorithms. Mostly using fearfull moments to find buy opportunities.

YouTube Caption Search: Search captions within your subscribed content.

Gmail Unsubscribe Finder: Easily locate and manage email subscriptions.

Custom Keyboard: Soldered my own modern design keyboard based on the QMK Firmware.

Home Lab: I am running a proxmox server with multiple VMs and docker containers to host projects, and a VPN to access them.

Skills : **Django, Python, Git, Celery, Airflow, API, Docker, Linux, HTML, CSS, SQL, PostgreSQL**

2014-2018 **System Engineer / Developer**, Metro Group

Managed IT infrastructure and provided support.

Automating User Management: Migrating from generic accounts to personal accounts, automating creation of user accounts and generating documentation.

Exchange Management: Managing generic mailboxes and a self-service tool.

Network Management: Managing LAN, Wi-Fi, Firewalls and setting up new stores.

Support: First and second line support.

Skills: **Networking: (DHCP, DNS, Exchange, Firewall), PowerShell Programming, Confluence**

2011-2014 **Sales Digital World**, Metro Group

Sales role focused on creating a great customer experience.

Sales: Listening to the needs of the customer and translating it to the correct product.

Skills: Sales, Communication

2012 **Teacher ICT**, GLTT

Breaking down information into small and easy to understand concepts.

2009-2010 **System Engineer / Development**, University Ghent

Managed IT infrastructure and developed a workflow tool.

Developing a workflow Tool: Writing a user story and developing a workflow tool for sales, laborants, customers and administration department.

IT Infrastructure Management: IT infrastructure management and support together with the central IT department.

Skills : **Web Development (.NET), SQL, HTML, CSS, Networking: (DHCP, DNS, Exchange, Firewall)**

Other Skills -

Languages: Mother Tongue in Dutch, very good English, very good French

Soft Skills: **Team player, eager to learn and teach, motivated, calm and**

thoughtful, creative, critical thinking, problem-solving, analytic.

Education 2005-2008 **Professional Bachelor ICT**, KaHo Sint-Lieven, Ghent, Belgium
1999-2005 **Industrial Sciences**, Don Bosco, Halle, Belgium

Hobbies Hiking w/wo tent, Renovating, DIY projects, Running, Mountain biking, Dancing, Exploring new Tech.